ML-KEM-768 Decapsulation Timing Assessment

Does decapsulation wall-clock timing reveal implicit rejection (FIPS 203 §6.3, Algorithm 18)? Run: 2026-11-07T15:11:25Z–15:14:52Z (2026-11-07 22:11:25–22:14:52 WIB). Harness: `src/run_timing.py`. Full machine-readable results: `outputs/timing-results.json` (2400 batches, 2,400,000 decapsulations). Unedited log: `outputs/run_timing-2026-11-07-fresh.log`. Digests: `outputs/SHA256SUMS`.

2400
batches × 1000 decapsulations

2,400,000
decapsulations

0
suspicious pairs

0
confirmation reruns

+0.400%
largest adversarial median difference (kyber-ref SYNTH-RAND)

+1.155%
Go null-control difference (noise floor)

Headline results

- **No timing difference between valid and implicitly-rejected decapsulations in any implementation.** Largest adversarial median difference: +0.400% (Kyber reference, SYNTH-RAND, neither test significant). Per implementation: pyca 0.124%, liboqs 0.110%, Go 0.220%, kyber-ref 0.400%. No pair met the rerun rule (|median diff| > 0.5% AND p < 0.01), so no confirmation reruns were performed.

- **The noise floor is the finding for Go.** Go's own VALID-vs-VALID null control differs by +1.496 µs (+1.155%, Mann-Whitney p = 0.0019) — larger than every Go adversarial effect. Sub-1% effects are undetectable for Go in this environment.
- **One discordant test pair (liboqs WRONGKEY):** Welch's t-test flags it (p = 0.0006) while the median difference is +0.017% and Mann-Whitney does not flag it (p = 0.2418) — a mean/outlier effect from a few slow batches, not a distributional shift.
- **FIPS 203 does not mandate constant-time execution** (the words "constant time" and "timing" do not appear in the standard); it requires the rejection flag to be destroyed, not returned (§3.3, §6.3). A null wall-clock result is therefore not a constant-time certification — and nothing here demonstrates exploitability either.
- Absolute speeds (VALID median): liboqs 22.953 µs, pyca 52.187 µs, Go 129.482 µs, Kyber reference 133.157 µs.

Implementations

ID	Implementation	Version	Key encoding
A	pyca/cryptography	50.0.2	ek 1184 B; dk as 64-byte seed (d‖z)
B	liboqs (ML-KEM-768, built from source; AVX2 path on this CPU)	0.12.0, commit f4b9622	ek 1184 B; sk 2400 B expanded
C	Go stdlib crypto/mlkem	go1.24.6	ek 1184 B; dk as 64-byte seed (d‖z)
D	pq-crystals/kyber Kyber768 reference (round-3, pre-FIPS 203 — lineage reference only)	commit 3edd5af	pk 1184 B; sk 2400 B

D implements round-3 Kyber768, not FIPS 203 ML-KEM: its shared secrets are not comparable with A/B/C. Only its timing behavior is compared. The measured liboqs path is the pqcrystals-kyber-standard AVX2 implementation, not the reference C code.

Method

5 fresh keypairs per implementation; per keypair, 6 input classes × 20 batches × 1000 decapsulations; batch order shuffled per round; 200 warm-up decaps per keypair (discarded). Timed region: pure decapsulation. Input classes: VALID (valid encapsulation), FLIP1 (ct[0] bit flipped), FLIPMID (ct[543] bit flipped), WRONGKEY (valid ciphertext, different keypair), SYNTH-ZERO (all-zero ciphertext), SYNTH-RAND (random ciphertext). All times are microseconds per decapsulation (batch means, n=100 per implementation per class). Pairwise tests: Welch's t-test and Mann-Whitney U (large-sample normal approximations), 95% CI for the mean difference, Cohen's d. Null control: VALID batches split in half, tested VALID-vs-VALID. Rerun rule: |median difference| > 0.5% of the VALID median AND p < 0.01 — no pair met it.

Timing matrix

A — pyca/cryptography 50.0.2 (VALID median 52.187 µs)

Class	n	Mean µs	Median µs	Stdev µs	p5 µs	p25 µs	p75 µs	p95 µs	p99 µs
VALID	100	53.319	52.187	3.393	51.800	51.973	52.802	62.487	68.258
FLIP1	100	53.806	52.230	5.156	51.795	51.959	53.109	62.420	78.347
FLIPMID	100	53.376	52.123	4.485	51.835	51.935	52.671	55.524	78.000
WRONGKEY	100	53.191	52.144	3.826	51.748	51.973	52.819	56.092	62.004
SYNTH-ZERO	100	53.275	52.149	5.464	51.834	51.991	52.580	56.028	62.368
SYNTH-RAND	100	53.525	52.165	4.058	51.777	51.981	53.169	58.528	72.944

Comparison	Median diff µs	Median diff %	Welch p	MWU p	Cohen's d	

						95% CI (mean diff) µs
VALID-vs-VALID (null control)	-0.142	-0.273%	0.5217	0.4649	-0.128	[-1.770, +0.898]
VALID-vs-FLIP1	+0.042	+0.081%	0.4303	0.7305	-0.112	[-1.697, +0.723]
VALID-vs-FLIPMID	-0.065	-0.124%	0.9194	0.53	-0.014	[-1.159, +1.045]
VALID-vs-WRONGKEY	-0.043	-0.082%	0.8036	0.9786	+0.035	[-0.875, +1.130]
VALID-vs-SYNTH-ZERO	-0.038	-0.073%	0.9457	0.8317	+0.010	[-1.217, +1.304]
VALID-vs-SYNTH-RAND	-0.022	-0.043%	0.6967	0.8489	-0.055	[-1.243, +0.831]

B — liboqs 0.12.0 (AVX2 path) (VALID median 22.953 µs)

Class	n	Mean µs	Median µs	Stdev µs	p5 µs	p25 µs	p75 µs	p95 µs	p99 µs
VALID	100	23.021	22.953	0.239	22.779	22.880	23.080	23.582	23.834
FLIP1	100	23.051	22.951	0.502	22.770	22.854	23.046	23.584	24.686
FLIPMID	100	23.199	22.955	2.235	22.769	22.863	23.047	23.410	23.804
WRONGKEY	100	23.215	22.957	0.509	22.801	22.856	23.362	24.276	24.430
SYNTH-ZERO	100	23.424	22.928	3.122	22.775	22.829	23.065	23.444	44.900
SYNTH-RAND	100	22.987	22.944	0.251	22.738	22.849	23.059	23.348	24.050

Comparison	Median diff µs	Median diff %	Welch p	MWU p	Cohen's d	

						95% CI (mean diff) µs
VALID-vs-VALID (null control)	-0.056	-0.244%	0.1279	0.1082	-0.305	[-0.166, +0.021]
VALID-vs-FLIP1	-0.002	-0.007%	0.596	0.511	-0.075	[-0.139, +0.080]
VALID-vs-FLIPMID	+0.002	+0.008%	0.4306	0.4606	-0.111	[-0.618, +0.263]
VALID-vs-WRONGKEY	+0.004	+0.017%	0.0005939	0.2418	-0.486	[-0.304, -0.083]
VALID-vs-SYNTH-ZERO	-0.025	-0.110%	0.1991	0.1907	-0.182	[-1.016, +0.212]
VALID-vs-SYNTH-RAND	-0.008	-0.036%	0.3154	0.2181	+0.142	[-0.033, +0.103]

C — Go 1.24.6 stdlib crypto/mlkem (VALID median 129.482 µs)

Class	n	Mean µs	Median µs	Stdev µs	p5 µs	p25 µs	p75 µs	p95 µs	p99 µs
VALID	100	130.401	129.482	4.064	128.141	128.517	130.868	135.513	136.585
FLIP1	100	131.002	129.215	5.415	128.043	128.645	130.716	139.059	158.419
FLIPMID	100	130.526	129.365	3.974	128.008	128.582	131.100	135.653	147.923
WRONGKEY	100	130.466	129.359	4.151	128.058	128.651	130.566	133.580	152.334
SYNTH-ZERO	100	131.797	129.766	6.980	128.147	128.783	131.430	143.475	165.757
SYNTH-RAND	100	130.595	129.530	3.304	128.060	128.859	131.057	140.125	142.379

Comparison				MWU p		

	Median diff µs	Median diff %	Welch p		Cohen's d	95% CI (mean diff) µs
VALID-vs-VALID (null control)	+1.496	+1.155%	0.01624	0.001921	+0.481	[+0.352, +3.464]
VALID-vs-FLIP1	-0.267	-0.206%	0.3748	0.8202	-0.126	[-1.928, +0.726]
VALID-vs-FLIPMID	-0.117	-0.090%	0.8263	0.6477	-0.031	[-1.239, +0.989]
VALID-vs-WRONGKEY	-0.123	-0.095%	0.9112	0.9883	-0.016	[-1.203, +1.074]
VALID-vs-SYNTH-ZERO	+0.284	+0.220%	0.08387	0.09859	-0.244	[-2.979, +0.187]
VALID-vs-SYNTH-RAND	+0.048	+0.037%	0.7116	0.2747	-0.052	[-1.220, +0.833]

D — pq-crystals/kyber Kyber768 reference (lineage reference, pre-FIPS 203) (VALID median 133.157 µs)

Class	n	Mean µs	Median µs	Stdev µs	p5 µs	p25 µs	p75 µs	p95 µs	p99 µs
VALID	100	134.868	133.157	5.986	131.519	132.457	134.649	143.508	162.835
FLIP1	100	134.677	133.225	5.197	131.714	132.434	134.897	140.259	164.155
FLIPMID	100	134.257	133.337	4.859	131.638	132.438	134.634	136.647	160.368
WRONGKEY	100	136.064	133.236	16.786	131.938	132.643	134.836	139.298	166.599
SYNTH-ZERO	100	134.534	133.222	5.184	131.717	132.493	134.511	137.754	161.157
SYNTH-RAND	100	134.951	133.690	5.390	131.665	132.728	135.004	144.078	162.277

Comparison	Median diff µs	Median diff %	Welch p	MWU p	Cohen's d	95% CI (mean diff) µs
VALID-vs-VALID (null control)	+0.226	+0.170%	0.4394	0.4607	-0.155	[-3.279, +1.424]
VALID-vs-FLIP1	+0.069	+0.051%	0.8094	0.6285	+0.034	[-1.363, +1.745]
VALID-vs-FLIPMID	+0.181	+0.136%	0.4277	0.8393	+0.112	[-0.900, +2.123]
VALID-vs-WRONGKEY	+0.079	+0.059%	0.5023	0.3188	-0.095	[-4.689, +2.297]
VALID-vs-SYNTH-ZERO	+0.066	+0.049%	0.6733	0.8508	+0.060	[-1.218, +1.886]
VALID-vs-SYNTH-RAND	+0.533	+0.400%	0.9176	0.1011	-0.015	[-1.662, +1.496]

Findings

- **F-1 — Informational (method):** the Go null control drifts more than any adversarial effect — the noise floor, not a leak. Go VALID-vs-VALID: +1.496 µs (+1.155%), MWU p = 0.0019; largest Go adversarial difference is SYNTH-ZERO at +0.220%.
- **F-2 — Informational:** liboqs WRONGKEY is discordant — Welch p = 0.0006 but median difference +0.017% and MWU p = 0.2418; a mean/outlier effect, which is why the protocol requires median + rank agreement.
- **F-3 — Positive:** no timing difference between valid and implicitly-rejected decapsulations in any implementation (20 pairs, largest median difference +0.400%, none suspicious).
- **F-4 — Informational:** absolute speeds differ widely; liboqs (AVX2) is fastest at 22.953 µs VALID median, ~5.6× faster than Go.
- **F-5 — Caveat:** a null wall-clock result is not proof of constant-time execution, and not proof of exploitability either. FIPS 203 does not mandate constant-time execution; the Kyber/ML-KEM reference lineage

has had real timing leaks despite constant-time-looking source (Clangover, CVE-2024-37880; KyberSlash).
- **F-6 — Informational:** constant-time structure verified in source for the Kyber reference (`verify()` + `cmov()` in `kem.c`); the liboqs AVX2 path actually measured was not source-verified here; pyca/OpenSSL constant-time status was not verified from a primary source in this run.

Full severity-ranked findings: `findings.md`. Claim-to-evidence index: `index.md`. Protocol and FIPS 203 requirements: `rubric.md`. Per-class statistics and tests: `matrix.md`.

Environment

`CPU: AMD EPYC 9D25 126-Core Processor, 4 vCPUs; Python 3.12.3; Linux-7.0.0-43-generic-x86_64-with-glibc2.39; perf_counter resolution 1e-09 s; load average at start 0.21, 0.29, 0.27; at end 1.05, 0.71, 0.45.`

An earlier partial run (2026-11-06, killed during kyber-ref keypair 1/5) is preserved unedited as `outputs/run_timing.log` (identical copy `outputs/run_timing-2026-11-06-partial.log`); no result in this pack uses it.